

PS Funktionale Programmierung, SS 2024 – Blatt 5

1. Write two procedures **checkProp** and **checkPropOneLine**. Both procedures shall take a list of items and a predicate, i.e. a procedure that returns a boolean value (e.g. `even?`). It shall return a new list that contains the corresponding boolean values that indicate whether or not the property holds for the respective element (i.e., if the predicate evaluates to `#t` or `#f`). The implementation (body) of **checkPropOneLine** shall be a single line of code and may use built-in procedures such as *range*, *filter*, *map*. **checkProp** must not use *range*, *filter*, *map*, or *checkPropOneLine*.

Example: `(checkProp even? (list 1 2 3 4))` → `'(#f #t #f #t)`

2. Write a procedure **doAll** that takes a list of procedures and an item as arguments and returns a list of the results of applying each procedure to the item.

Examples:

`(doAll (list sqrt square cube) 4)` → `'(2 16 64)`

`(doAll (list length car cdr) (list 1 2 3))` → `'(3 1 (2 3))`

3.

a. Given a list of numbers that represent temperature values in Fahrenheit, write a procedure **toCelsius** that converts this list into Celsius scale. Use **map** and **λ-Expressions**.

Example: `(toCelsius (list -40 32 50))` → `'(-40 0 10)`

b. Write a procedure **checkAll** that shall take a list of items and a predicate, i.e. a procedure that returns a boolean value (e.g. `even?`). It shall return false in case the list contains at least one element where the given property does not hold (i.e., if the predicate evaluates `#f`); otherwise true. You must use **foldr**.

4. You are allowed to use any procedure you want:

a. Write a procedure **countEvenGreater20** that takes a list of values and returns the number of values that are even and greater than 20.

Example: `(countEvenGreater20 (list 1 4 22 24 26 33 -44))` → 3

b. Write a procedure **sumEvenGreater20** takes a list of values and returns the sum of the values that are even and greater than 20.

Example: `(sumEvenGreater20 (list 1 4 22 24 26 33 -44))` → 72

5. Use the conventional interfaces (e.g., `map`, `filter`, `foldr`, ...) discussed in the lecture and lambda expressions to calculate the following: $\sum_{i=10}^{100} i^3 \mid i \% 2 == 0$